Network Analysis Part 2 Solutions

2 October 2016 by Miodrag Sljukic (http://www.r-exercises.com/author/miodrag/)

Difficulty level: ★★⯪☆☆ 2.33/5 (3 votes)

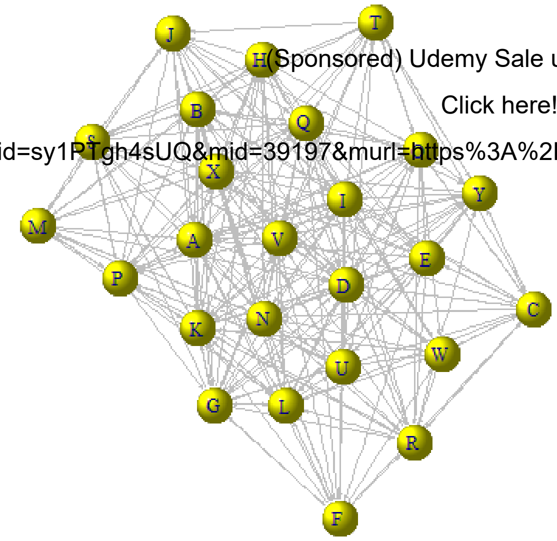
Below are the solutions to these (http://www.r-exercises.com/2016/10/02/network-analysis-part-2-exercises/) exercises, Network Analysis Part 2.

```
####################
#                  #
#    Exercise 1    #
#                  #
####################

d <- read.csv("sociogram-employees-un.csv", header=FALSE)
g <- graph.adjacency(as.matrix(d), mode="directed")
V(g)$name <- LETTERS[1:NCOL(d)]
V(g)$color <- "yellow"
V(g)$shape <- "sphere"
E(g)$color <- "gray"
E(g)$arrow.size <- 0.2

####################
#                  #
#    Exercise 2    #
#                  #
####################

plot(g)
```




```
####################
#                  #
#    Exercise 3    #
#                  #
####################

diameter(g)
```

```
## [1] 3
```

```
mean(closeness(g))
```

```
## [1] 0.02706843
```

```
####################
#                  #
#    Exercise 4    #
#                  #
####################

mean(betweenness(g))
```

```
## [1] 13.04
```

```
####################
#                  #
#    Exercise 5    #
#                  #
####################

graph.density(g)
```

```
## [1] 0.4616667
```



```
mean(degree(g, mode="all"))
```

```
## [1] 22.16
```



```
####################
#                  #
#    Exercise 6    #
#                  #
####################

reciprocity(g)
```

```
## [1] 0.4837545
```

```
mean(transitivity(g))
```

```
## [1] 0.7145809
```

```
####################
#                  #
#    Exercise 7    #
#                  #
####################

mean(eccentricity(g))
```

```
## [1] 2
```

```
mean_distance(g)
```

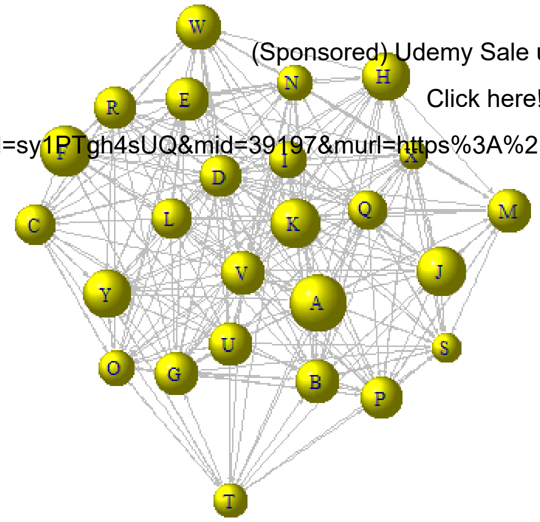
```
## [1] 1.543333
```

```
####################
#                  #
#    Exercise 8    #
#                  #
####################

hs <- hub.score(g)$vector
hs
```

```
##         A         B         C         D         E         F         G 
## 1.0000000 0.7476601 0.6988751 0.7321069 0.7584761 0.9014230 0.7648633 
##         H         I         J         K         L         M         N 
## 0.8352513 0.6446838 0.8695903 0.8732329 0.7025755 0.7595805 0.6295569 
##         O         P         Q         R         S         T         U 
## 0.6345403 0.7346497 0.6711406 0.7402970 0.5313184 0.6036291 0.7473912 
##         V         W         X         Y 
## 0.7641856 0.7808829 0.5071989 0.8309624
```

```
plot(g, layout=layout.fruchterman.reingold, vertex.size=hs*25)
```




```
which.max(hs)
```

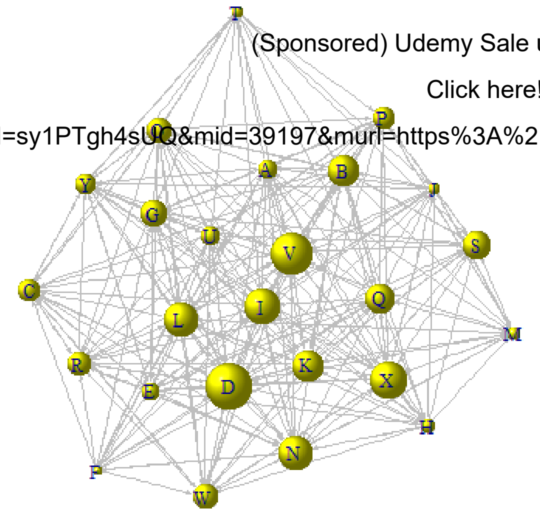
```
## A
## 1
```

```
####################
#                  #
#    Exercise 9    #
#                  #
####################

as <- authority.score(g)$vector
as
```

```
##         A         B         C         D         E         F         G
## 0.4132531 0.6786441 0.4879129 1.0000000 0.3945946 0.1903974 0.5622948
##         H         I         J         K         L         M         N
## 0.3266580 0.7547999 0.2660651 0.6868545 0.7368024 0.2838372 0.7452176
##         O         P         Q         R         S         T         U
## 0.5492443 0.4906222 0.6437024 0.5103429 0.5966113 0.2532414 0.4226322
##         V         W         X         Y
## 0.8902226 0.5279042 0.8052346 0.4475643
```

```
plot(g, layout=layout_nicely, vertex.size=as*20)
```




```
which.max(as)
```

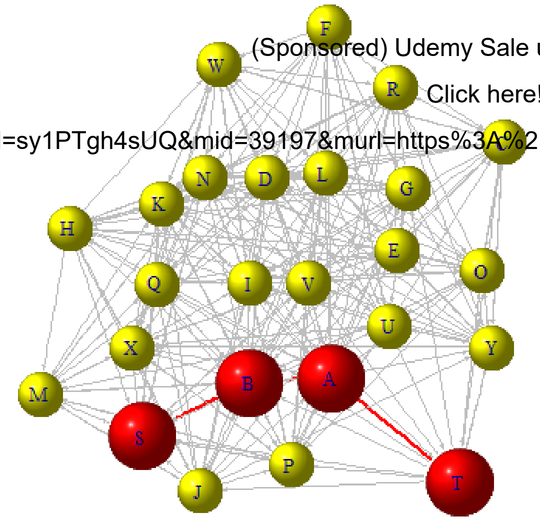
```
## D
## 4
```

```
####################
#                  #
#    Exercise 10   #
#                  #
####################

diameter.nodes <- get.diameter(g)
diameter.nodes
```

```
## + 4/25 vertices, named:
## [1] S B A T
```

```
V(g)$size <- 20
V(g)[diameter.nodes]$color <- "red"
V(g)[diameter.nodes]$size <- V(g)[diameter.nodes]$size+10
E(g)$width <- 1
E(g, path=diameter.nodes)$color <- "red"
E(g, path=diameter.nodes)$width <- 2
plot.igraph(g, layout=layout.fruchterman.reingold)
```



[(http://www.r-exercises.com/2016/10/02/network-analysis-part-2-solutions/?print=pdf)](http://www.r-exercises.com/2016/10/02/network-analysis-part-2-solutions/?print=pdf)

[(http://www.r-exercises.com/2016/10/02/network-analysis-part-2-solutions/?print=print)](http://www.r-exercises.com/2016/10/02/network-analysis-part-2-solutions/?print=print)

What's next:

- Explore all our (>1000) R exercises (http://www.r-exercises.com/start-here-to-learn-r/)
- Find an R course using our R Course Finder (http://www.r-exercises.com/r-courses/) directory
- Subscribe (http://www.r-exercises.com/subcribe/) to receive weekly updates and bonus sets by email
- Share with your friends and colleagues using the buttons below

Please help us to improve R-exercises:

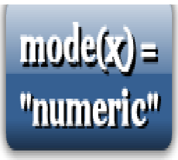
Did you like this set? (1 star = no, not at all; 5 stars = yes, very much!) ☆☆☆☆☆

How difficult was this for you? (1 star=easy, 10 stars=hard) ☆☆☆☆☆☆☆☆☆☆

Are you a beginner (1 star), intermediate (2 stars) or advanced (3 stars) R user? ☆☆☆

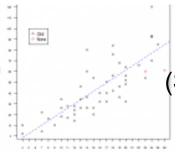
0 (/#facebook) (/#twitter) (/#google_plus)
0 (/#linkedin) (/#reddit) (/#hacker_news)
(/#whatsapp) (/#email)
0
(https://www.addtoany.com/share#url=http%3A%2F%2Fwww.r-exercises.com%2F2016%2F10%2F02%2Fnetwork-analysis-part-2-solutions%2F&title=Network%20Analysis%20Part%202%20Solutions)

Related Exercise Sets:



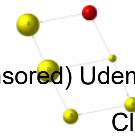
Mode exercises

(http://www.r-exercises.com/2016/02/14/mode-exercises/)



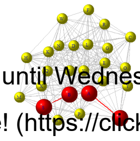
Advanced Base Graphics Exercises

(http://www.r-exercises.com/2016/09/23/advanced-base-graphics-exercises/)



Network Analysis Part 1 Exercises

(http://www.r-exercises.com/2016/09/15/network-analysis-part-1-exercises/)



Network Analysis Part 2 Exercises

(http://www.r-exercises.com/2016/10/02/network-analysis-part-2-exercises/)

Network Analysis Part 3 Exercises

(http://www.r-exercises.com/2016/10/20/network-analysis-part-3-exercises/)

About Miodrag Sljukic

I have master degree in finance and a post-graduate specialization in e-business. I am the founder of Serbian statistical portal e-Statistika (http://www.e-statistika.rs). My principal activities include statistical analysis, software development and trading shares.